

UNITED STATES SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

**ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES
EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2024

VERDANT SOLARIS INDUSTRIES, INC.

(Exact name of registrant as specified in its charter)

Commission File Number: 001-38479

Delaware (State of Incorporation) — IRS Employer ID No. 84-2173694

4801 Heliostat Way, Tempe, Arizona 85284

Telephone: (480) 555-7100

Common Stock, par value \$0.001 per share — NYSE: VRSL

Table of Contents

	Item	Description	Page
PART I	1	Business	3
	1A	Risk Factors	5
	1B	Unresolved Staff Comments	7
	2	Properties	7
	3	Legal Proceedings	8
PART II	5	Market for Registrant's Common Equity	9
	6	[Reserved]	9
	7	Management's Discussion and Analysis	10
	7A	Quantitative & Qualitative Disclosures About Market Risk	13
	8	Financial Statements and Supplementary Data	14
		Consolidated Statements of Income	14
		Consolidated Balance Sheets	15
		Consolidated Statements of Cash Flows	16
		Consolidated Statements of Stockholders' Equity	17
		Note 1 – Significant Accounting Policies	18
		Note 2 – Revenue Recognition	18
		Note 3 – Earnings per Share	19
		Note 4 – Acquisitions	19
		Note 5 – Goodwill and Other Intangible Assets	20
		Note 6 – Inventories	20
		Note 7 – Property, Plant and Equipment	21
		Note 8 – Long-Term Debt	22
		Note 9 – Income Taxes	23
		Note 10 – Leases	24
		Note 11 – Employee Benefits	24
		Note 12 – Stock-Based Compensation	25
		Note 13 – Commitments and Contingencies	25
		Note 14 – Segment Information	26
		Note 15 – Geographic Information	27
		Note 16 – Selected Quarterly Financial Data (Unaudited)	28
		Note 17 – Subsequent Events	28

PART I

Item 1. Business

Verdant Solaris Industries, Inc. ("Verdant Solaris," "VSI," or the "Company") is a vertically integrated renewable-energy company that designs, manufactures, and deploys photovoltaic modules, utility-scale battery energy storage systems, grid integration and software services, and originates and develops large-scale renewable-energy projects globally. The Company organizes its business into four reportable operating segments: Photovoltaic Manufacturing ("PVM"), Energy Storage Systems ("ESS"), Grid Integration Services ("GIS"), and Renewable Project Development ("RPD"). A fifth reporting bucket, Corporate / Other, captures unallocated corporate function expenses, equity-method investments, intercompany eliminations, and centrally held cash, restricted cash and investments.

The Company was incorporated in Delaware on March 14, 2006 under the name 'SunBridge Photonics, Inc.' and was renamed Verdant Solaris Industries, Inc. on August 1, 2014 following the acquisition of Arc-Reactor Storage Holdings, LLC. The Company completed its initial public offering on May 23, 2018, listing its common stock on the New York Stock Exchange under the symbol "VRSL". As of December 31, 2024 the Company had approximately 18,427 full-time employees worldwide (December 31, 2023: 16,892; December 31, 2022: 14,738).

Segment Snapshot

The Photovoltaic Manufacturing segment operated 11 production facilities in 2024 with combined nameplate capacity of 18.4 GW of cadmium-telluride and tandem-perovskite modules. The Energy Storage Systems segment shipped 14.7 GWh of integrated battery systems during the year. The Grid Integration Services segment provides hardware-agnostic control software, asset management, and field engineering, and reported a backlog of \$1,247.3 million at year-end. The Renewable Project Development segment originated 3.2 GW of new project capacity and completed sale-and-monetization transactions for 1.8 GW of assets.

Item 1A. Risk Factors

Investing in our common stock involves a high degree of risk. The following risk factors, together with the other information contained in this Annual Report on Form 10-K, should be carefully considered. Additional risks not currently known to us or that we currently consider immaterial may also impair our business operations.

Risks Related to Capital Intensity

Our business is highly capital intensive. We expect to incur substantial capital expenditures over the next five years to expand cell and module capacity in Arizona, Texas, and Saxony, to construct two new lithium iron phosphate cell facilities, and to scale our integrated software platform. We currently project aggregate capital expenditures in the range of \$5.4 billion to \$6.1 billion over fiscal years 2025 through 2029. A delay or cost overrun in any of these projects could materially adversely affect our financial condition. Our 2024 consolidated capital expenditures of \$1,284.9 million represented 14.4% of consolidated net sales. The Company's largest single capital project in 2024 was the Phase III expansion of the Tempe Module Manufacturing Facility, which incurred \$317.4 million of capitalized cost during the year and is expected to incur an additional \$186.2 million in 2025.

Commodity and Supply-Chain Risk

The Company is exposed to commodity-price volatility for tellurium, lithium, copper, aluminum, polysilicon, and electricity used in cell production. During 2024 the Company's spot purchases of refined tellurium totaled 142.7 metric tons at an average price of \$487 per kilogram, compared with 121.4 metric tons at \$412 per kilogram in 2023.

Item 1B. Unresolved Staff Comments

None.

Item 2. Properties

Facility	Location	Segment	Sq. Ft. (thousands)	Owned / Leased
Tempe Module Mfg. Facility	Tempe, AZ	PVM	1,247.4	Owned
Phoenix Tandem Cell Pilot Plant	Phoenix, AZ	PVM	318.2	Owned
Greenville Module Plant	Greenville, OH	PVM	892.6	Owned
Austin Cell Facility	Austin, TX	PVM	748.1	Owned
Saxony Module Plant	Dresden, Germany	PVM	612.7	Owned
Reno Gigafactory (Battery)	Reno, NV	ESS	1,887.4	Owned
McAllen Battery Pack Plant	McAllen, TX	ESS	543.2	Leased
Stuttgart Battery Engineering Ctr.	Stuttgart, Germany	ESS	127.8	Leased
Boulder Software & Operations Ctr.	Boulder, CO	GIS	248.7	Leased
Houston Project Dev. Office	Houston, TX	RPD	84.3	Leased
Tempe Corporate HQ	Tempe, AZ	Corporate	192.4	Owned

In addition to the major facilities listed above, the Company leases sales offices and field engineering depots in 27 countries totaling approximately 487 thousand square feet. The Company believes its properties are adequate for current and near-term needs.

Item 3. Legal Proceedings

The Company is involved in routine litigation incidental to its business. On September 4, 2023, plaintiffs in the matter Helios Capital Partners v. Verdant Solaris Industries, Inc. filed a putative securities class action in the U.S. District Court for the District of Arizona alleging violations of Section 10(b) of the Exchange Act. The complaint seeks unspecified damages. The Company believes the claims are without merit and intends to vigorously defend itself; the matter is in early stages and a reasonable estimate of loss, if any, cannot be made at this time. The Company has recorded no accrual related to this matter as of December 31, 2024.

PART II

Item 5. Market for Registrant's Common Equity

The Company's common stock has traded on the New York Stock Exchange under the symbol 'VRSL' since May 23, 2018. As of February 14, 2025, there were 437 holders of record. The Company has not paid cash dividends on its common stock and does not currently intend to do so. During the year ended December 31, 2024, the Company repurchased 4,287,142 shares of its common stock under its publicly announced repurchase program at an average price of \$74.83 per share, for a total cost of \$320.8 million. As of December 31, 2024, \$679.2 million remained available for repurchase under the program.

Item 6. [Reserved]

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations

Overview

Fiscal year 2024 was the seventh consecutive year of double-digit revenue growth for Verdant Solaris. Consolidated net sales increased 14.9% year-over-year to \$8,247.2 million (2023: \$6,995.8 million; 2022: \$5,991.0 million). Gross profit increased to \$2,247.4 million (2023: \$1,884.2 million), and gross margin expanded to 27.3% (2023: 26.9%). Operating income rose 21.7% to \$948.3 million (2023: \$779.0 million). Net income attributable to Verdant Solaris common shareholders was \$621.4 million (2023: \$497.6 million) and diluted earnings per share was \$6.84 (2023: \$5.47).

Liquidity and Capital Resources

Net cash provided by operating activities was \$1,486.7 million in 2024, compared with \$1,247.3 million in 2023 and \$1,089.4 million in 2022. The Company invested \$1,284.9 million in property, plant and equipment in 2024, of which \$1,257.8 million related to the four reportable segments and \$27.1 million related to corporate facilities. The largest single investment was the Phase III expansion of the Tempe module-manufacturing facility within the Photovoltaic Manufacturing segment, which represented \$317.4 million of 2024 spend (2023: \$284.7 million). We also spent \$401.7 million expanding cell and pack production capacity at the Reno gigafactory within the Energy Storage Systems segment.

On May 17, 2024, the Company issued \$750.0 million aggregate principal amount of 5.125% Senior Notes due 2034 in a registered public offering. Net proceeds, after deducting underwriting discounts and offering expenses of \$6.8 million, were \$743.2 million. The proceeds were used to repay the \$400.0 million outstanding balance under the Company's Revolving Credit Facility and for general corporate purposes.

Segment Results Summary

Segment	Net Sales (USD millions)			% change	
	2024	2023	2022	2024 vs 2023	2023 vs 2022
Photovoltaic Manufacturing	3,812.5	3,284.7	2,891.2	16.1%	13.6%
Energy Storage Systems	2,891.4	2,367.1	1,923.6	22.2%	23.1%
Grid Integration Services	854.6	742.8	651.9	15.0%	13.9%
Renewable Project Development	688.7	601.2	524.3	14.6%	14.7%
Consolidated net sales	8,247.2	6,995.8	5,991.0	17.9%	16.8%

Photovoltaic Manufacturing net sales grew on increased module shipments (16.8 GW in 2024 vs. 14.1 GW in 2023) partially offset by a 4.2% decline in blended average selling price. Energy Storage Systems revenue growth reflected commissioning of 5.4 GWh of contracted backlog at higher per-MWh pricing under contracts signed in late 2022 and early 2023.

Critical Accounting Estimates

The Company's critical accounting estimates include revenue recognition for construction-type contracts using cost-to-cost percentage-of-completion, valuation of inventories at lower of cost or net realizable value, useful lives of property, plant and equipment, impairment testing of long-lived assets and goodwill, and income tax positions including realization of deferred tax assets and uncertain tax positions.

Item 7A. Quantitative and Qualitative Disclosures About Market Risk

The Company is exposed to interest rate risk on its variable-rate revolving credit facility and on its short-term investments. A hypothetical 100 basis-point increase in short-term interest rates would have decreased the Company's pretax income for 2024 by approximately \$4.1 million, holding all else constant.

The Company is exposed to foreign-currency risk primarily through its European operations (Euro and British Pound) and its Japan operations (Japanese Yen). Approximately 33.6% of consolidated 2024 revenue, or \$2,770.5 million, was denominated in currencies other than the U.S. dollar (2023: 32.5%, or \$2,272.9 million).

Forward currency contracts outstanding at December 31, 2024 had a notional value of \$1,478.3 million and an aggregate fair value of \$14.7 million reflected within Other current assets. Commodity hedges (primarily aluminum and copper) outstanding at December 31, 2024 had notional values of \$187.2 million.

Item 8. Financial Statements and Supplementary Data

Consolidated Statements of Income

(USD millions, except per-share data)	2024	2023	2022
Net sales	8,247.2	6,995.8	5,991.0
Cost of sales	(6,000.0)	(5,111.6)	(4,408.4)
Gross profit	2,247.2	1,884.2	1,582.6
Selling, general and administrative	(728.4)	(642.7)	(568.2)
Research and development	(412.6)	(365.8)	(314.7)
Restructuring and other charges	(158.1)	(96.7)	(84.3)
Operating income	948.1	779.0	615.4
Interest expense	(82.6)	(71.4)	(64.8)
Interest income	31.4	22.8	18.1
Other income (expense), net	8.7	12.6	7.4
Income before income taxes	905.6	743.0	576.1
Provision for income taxes	(206.2)	(165.6)	(146.4)
Net income attributable to noncontrolling interests	(8.0)	(7.2)	(6.1)
Net income attributable to common stockholders	691.4	570.2	423.6
Earnings per share — basic	6.91	5.53	4.71
Earnings per share — diluted	6.84	5.47	4.66
Weighted-average shares — basic (millions)	89.9	90.0	89.7
Weighted-average shares — diluted (millions)	90.8	90.9	90.6

Consolidated Balance Sheets

(USD millions)	Dec 31, 2024	Dec 31, 2023
ASSETS		
Current assets:		
Cash and cash equivalents	1,047.3	892.6
Short-term investments	248.7	187.4
Accounts receivable, net of allowance (\$23.1; \$19.8)	1,387.4	1,178.6
Inventories	1,892.4	1,612.3
Contract assets	287.6	248.7
Other current assets	184.2	162.3
Total current assets	5,047.6	4,281.9
Property, plant and equipment, net	4,182.7	3,612.5
Operating lease right-of-use assets	287.4	248.6
Goodwill	642.3	598.7
Other intangible assets, net	287.6	324.1
Deferred tax assets, net	168.4	184.2
Other long-term assets	232.6	187.4
Total assets	10,848.6	9,437.4
LIABILITIES AND STOCKHOLDERS' EQUITY		
Current liabilities:		
Accounts payable	987.4	823.6
Accrued liabilities	612.8	524.7
Current portion of long-term debt	127.4	84.2
Contract liabilities	428.6	362.4
Other current liabilities	184.7	168.3
Total current liabilities	2,340.9	1,963.2
Long-term debt	2,187.4	1,612.8
Operating lease liabilities, non-current	264.3	228.7
Deferred tax liabilities	184.7	162.3
Other long-term liabilities	287.4	248.6
Total liabilities	5,264.7	4,215.6
Stockholders' equity:		
Common stock, \$0.001 par; 89.7 (2024); 90.1 (2023) shares outstanding	0.1	0.1
Additional paid-in capital	2,124.7	2,087.4
Retained earnings	3,487.4	3,186.8
Accumulated other comprehensive loss	(78.4)	(82.6)
Treasury stock, at cost	(132.4)	—
Total stockholders' equity attributable to Verdant Solaris	5,401.4	5,191.7
Noncontrolling interests	182.5	30.1
Total stockholders' equity	5,583.9	5,221.8
Total liabilities and stockholders' equity	10,848.6	9,437.4

Consolidated Statements of Cash Flows

(USD millions)	2024	2023	2022
Cash flows from operating activities:			
Net income (including noncontrolling interests)	629.4	504.8	418.0
Adjustments to reconcile net income to net cash from operating activities:			
Depreciation and amortization	621.5	523.9	444.3
Stock-based compensation expense	118.4	98.7	84.2
Deferred income taxes	(28.7)	(34.2)	(18.9)
Loss (gain) on disposal of assets	12.4	(8.7)	(4.3)
Inventory write-downs	31.7	24.6	21.4
Changes in operating assets and liabilities:			
Accounts receivable	(208.8)	(187.4)	(241.6)
Inventories	(280.1)	(247.3)	(184.7)
Contract assets	(38.9)	(46.2)	(38.1)
Accounts payable	163.8	187.6	194.3
Accrued liabilities	88.1	62.4	78.5
Contract liabilities	66.2	84.7	62.4
Other, net	311.7	284.4	273.7
Net cash provided by operating activities	1,486.7	1,247.3	1,089.4
Cash flows from investing activities:			
Purchases of property, plant and equipment	(1,284.9)	(1,082.4)	(921.4)
Proceeds from disposal of property, plant and equipment	12.4	8.7	6.4
Purchases of marketable securities	(421.3)	(312.6)	(287.4)
Proceeds from sales and maturities of marketable securities	362.7	324.8	248.6
Acquisitions, net of cash acquired	(48.7)	—	(127.4)
Investment in unconsolidated affiliates	(28.4)	(18.6)	(12.7)
Other, net	8.2	(4.1)	3.2
Net cash used in investing activities	(1,400.0)	(1,084.2)	(1,090.7)
Cash flows from financing activities:			
Proceeds from issuance of long-term debt	750.0	—	400.0
Repayments of long-term debt	(127.4)	(84.2)	(78.6)
Proceeds (repayments) from revolving credit facility, net	(400.0)	200.0	100.0
Repurchases of common stock	(320.8)	(184.6)	(147.3)
Proceeds from stock option exercises	48.7	62.4	41.3
Tax withholdings on stock-based awards	(38.4)	(31.7)	(24.6)
Distributions to noncontrolling interests	(7.2)	(5.8)	(4.7)
Other, net	(4.3)	(2.8)	(1.7)
Net cash provided by (used in) financing activities	(99.4)	(46.7)	284.4
Effect of exchange-rate changes on cash	(12.6)	8.4	(15.2)

Net increase (decrease) in cash and cash equivalents	(25.3)	124.8	267.9
Cash and cash equivalents at beginning of period	1,072.6	947.8	679.9
Cash and cash equivalents at end of period	1,047.3	1,072.6	947.8

Consolidated Statements of Stockholders' Equity

(USD millions)	Common Stock	Add'l Paid-In Capital	Retained Earnings	AOCI	Treasury Stock	NCI	Total
Balance, Dec 31, 2021	0.1	1,984.7	2,403.6	(72.4)	—	27.4	4,343.4
Net income	—	—	411.9	—	—	6.1	418.0
Other comprehensive loss	—	—	—	(8.7)	—	—	(8.7)
Stock-based compensation	—	84.2	—	—	—	—	84.2
Share repurchases	—	—	(147.3)	—	—	—	(147.3)
Stock issuances under plans	—	16.7	—	—	—	—	16.7
Distributions to NCI	—	—	—	—	—	(4.7)	(4.7)
Balance, Dec 31, 2022	0.1	2,085.6	2,668.2	(81.1)	—	28.8	4,701.6
Net income	—	—	497.6	—	—	7.2	504.8
Other comprehensive income	—	—	—	(1.5)	—	—	(1.5)
Stock-based compensation	—	98.7	—	—	—	—	98.7
Share repurchases	—	—	(184.6)	—	—	—	(184.6)
Stock issuances under plans	—	30.7	—	—	—	—	30.7
Distributions to NCI	—	—	—	—	—	(5.8)	(5.8)
Acquisition of NCI	—	(127.6)	5.6	—	—	(0.1)	(122.1)
Balance, Dec 31, 2023	0.1	2,087.4	2,986.8	(82.6)	—	30.1	5,021.8
Net income	—	—	621.4	—	—	8.0	629.4
Other comprehensive income	—	—	—	4.2	—	—	4.2
Stock-based compensation	—	118.4	—	—	—	—	118.4
Share repurchases	—	—	(120.8)	—	(200.0)	—	(320.8)
Stock issuances under plans	—	10.3	—	—	—	—	10.3
Stock retirements	—	(91.4)	—	—	67.6	—	(23.8)
Distributions to NCI	—	—	—	—	—	(7.2)	(7.2)
Acquisition / contribution from NCI	—	—	—	—	—	151.6	151.6
Balance, Dec 31, 2024	0.1	2,124.7	3,487.4	(78.4)	(132.4)	182.5	5,583.9

Notes to the Consolidated Financial Statements

Note 1. Summary of Significant Accounting Policies

The Consolidated Financial Statements are prepared in accordance with U.S. GAAP. Use of estimates is required, including for revenue recognition, inventory obsolescence, useful lives, impairment of long-lived assets, valuation of share-based awards, and income taxes. Functional currency for foreign subsidiaries is generally the local currency. Translation adjustments are recorded within AOCI.

Note 2. Revenue Recognition

The Company recognizes revenue when control of the promised good or service is transferred to the customer. For module sales, revenue is recognized at a point in time, generally upon shipment. For project-development contracts and integrated energy-storage system installations, revenue is recognized over time using the cost-to-cost percentage-of-completion method. The following table presents the Company's revenues disaggregated by recognition pattern.

(USD millions)	2024	2023	2022
Point-in-time	4,891.6	4,218.7	3,684.3
Over time	3,355.6	2,777.1	2,306.7
Total net sales	8,247.2	6,995.8	5,991.0

Note 3. Earnings per Share

Basic earnings per share is computed by dividing net income attributable to common stockholders by the weighted-average number of common shares outstanding during the period. Diluted EPS reflects the potential dilution from stock options, restricted stock units, and performance share units.

Note 4. Acquisitions

On August 12, 2024, the Company acquired 80% of HelioGrid Analytics GmbH, a Munich-based grid-software firm, for \$48.7 million in cash, net of acquired cash. The acquisition was accounted for as a business combination. The purchase price was allocated \$14.2 million to tangible net assets, \$42.3 million to identifiable intangible assets (primarily developed technology and customer relationships, with weighted-average useful lives of 7.4 years), \$11.4 million to goodwill (assigned to the GIS segment), and \$19.2 million to the noncontrolling interest. No acquisitions were completed during the year ended December 31, 2023. During 2022 the Company acquired 100% of NorthStar Renewables, LLC for \$127.4 million.

Note 5. Goodwill and Other Intangible Assets

(USD millions)	PVM	ESS	GIS	RPD	Total
Balance, December 31, 2022	187.4	234.7	104.6	62.8	589.5
Foreign currency translation	1.2	0.4	0.8	—	2.4
Acquisitions	—	—	—	—	—
Adjustments to prior acquisitions	—	5.3	—	1.5	6.8
Balance, December 31, 2023	188.6	240.4	105.4	64.3	598.7
Foreign currency translation	—	(0.2)	(0.5)	—	(0.7)
Acquisitions	—	—	11.4	—	11.4
Adjustments to prior acquisitions	12.2	8.4	—	12.3	32.9
Balance, December 31, 2024	200.8	248.6	116.3	76.6	642.3

Note 6. Inventories

(USD millions)	Dec 31, 2024	Dec 31, 2023
Raw materials	421.7	354.2
Work in process	612.3	528.9
Finished goods	858.4	729.2
Total inventories	1,892.4	1,612.3

Note 7. Property, Plant and Equipment

(USD millions)	Dec 31, 2024	Dec 31, 2023
Land and land improvements	234.7	212.8
Buildings and improvements	1,847.6	1,648.7
Manufacturing machinery and equipment	4,287.4	3,602.9
Software	187.4	168.3
Vehicles and other	84.6	72.4
Construction in progress	612.3	487.6
Property, plant and equipment, gross	7,254.0	6,192.7
Less: accumulated depreciation	(3,071.3)	(2,580.2)
Property, plant and equipment, net	4,182.7	3,612.5

Additions to property, plant and equipment (gross, including capitalized interest of \$10.4 million in 2024 and \$9.1 million in 2023, and capitalized software development costs) were \$1,301.6 million in 2024, \$1,095.3 million in 2023, and \$945.0 million in 2022. These amounts differ from cash paid for capital expenditures presented in the Consolidated Statements of Cash Flows due to timing of vendor payments and from segment capital expenditures presented in Note 14 (which exclude capitalized interest and certain corporate software additions). Depreciation expense for the years ended December 31, 2024, 2023 and 2022 was \$491.1 million, \$412.7 million, and \$354.6 million, respectively.

Note 8. Long-Term Debt

(USD millions)	Dec 31, 2024	Dec 31, 2023
4.250% Senior Notes due 2027	500.0	500.0
4.875% Senior Notes due 2030	600.0	600.0
5.125% Senior Notes due 2034	750.0	—
Term Loan A (variable, SOFR + 1.50%)	350.0	397.4
Revolving Credit Facility (\$1.0B capacity)	—	200.0
Equipment financing notes	127.4	84.2
Other borrowings	12.7	8.6
Less: unamortized discounts and issuance costs	(25.3)	(93.2)
Total debt	2,314.8	1,697.0
Less: current portion	(127.4)	(84.2)
Long-term debt, net	2,187.4	1,612.8

Scheduled aggregate maturities of long-term debt as of December 31, 2024 are: \$127.4 million (2025), \$187.6 million (2026), \$562.4 million (2027), \$112.7 million (2028), \$74.6 million (2029), and \$1,275.4 million thereafter.

Note 9. Income Taxes

The components of the provision for income taxes consist of the following:

(USD millions)	2024	2023	2022
Current — Federal	187.6	142.4	118.3
Current — State	32.1	24.7	21.4
Current — Foreign	15.2	12.3	10.5
Deferred — Federal	(21.4)	(8.7)	(4.2)
Deferred — State	(4.8)	(1.9)	(0.6)
Deferred — Foreign	(2.5)	(3.2)	1.0
Total provision for income taxes	206.2	165.6	146.4

Note 10. Leases

The Company has operating leases for office space, warehouses, sales offices, and certain manufacturing equipment. The weighted-average remaining lease term for operating leases is 6.4 years (2023: 6.7 years), and the weighted-average discount rate is 4.8% (2023: 4.5%). Operating-lease cost for 2024 was \$58.7 million (2023: \$51.4 million; 2022: \$46.2 million). Future minimum lease payments are: \$64.2 million (2025), \$58.7 million (2026), \$51.4 million (2027), \$44.2 million (2028), \$37.6 million (2029), and \$98.7 million thereafter.

Note 11. Employee Benefits

The Company sponsors a 401(k) plan for substantially all U.S. employees with a Company match of 50% on the first 6% of eligible compensation. Company contributions to the 401(k) plan were \$32.1 million in 2024 (2023: \$27.4 million; 2022: \$24.1 million). The Company also sponsors defined-benefit pension plans in Germany and Japan covering certain eligible employees. The aggregate projected benefit obligation of these plans at December 31, 2024 was \$187.4 million (2023: \$162.3 million) with plan assets of \$142.6 million (2023: \$128.7 million), resulting in a funded status of \$(44.8) million at December 31, 2024.

Note 12. Stock-Based Compensation

Stock-based compensation expense was \$118.4 million in 2024 (2023: \$98.7 million; 2022: \$84.2 million). At December 31, 2024 there was \$187.6 million of unrecognized compensation expense related to outstanding awards, expected to be recognized over a weighted-average remaining period of 2.4 years.

Note 13. Commitments and Contingencies

The Company had purchase obligations of \$1,847.6 million as of December 31, 2024, primarily related to long-term polysilicon, tellurium, lithium carbonate, and electricity supply agreements. The Company has outstanding letters of credit of \$148.7 million. The Company is involved in various legal matters arising in the ordinary course of business; refer to Item 3 above for a description of material matters.

Note 14. Segment Information

The Company's reportable operating segments are: Photovoltaic Manufacturing ("PVM"), Energy Storage Systems ("ESS"), Grid Integration Services ("GIS"), and Renewable Project Development ("RPD"). The chief operating decision maker (the Chief Executive Officer) evaluates segment performance based on segment net sales and segment operating income. Corporate / Other consists of unallocated corporate function expense, equity-method investments, foreign-currency hedging gains and losses, intercompany eliminations, and the centrally held cash and investments portfolio. Segment capital expenditures presented below represent additions to property, plant and equipment directly attributable to each segment and exclude capitalized interest and certain corporate-level software investments included in consolidated additions disclosed in Note 7.

(a) Net Sales by Reportable Segment

Net Sales (USD millions)	2024	2023	2022	2021	2020
Photovoltaic Manufacturing	3,812.5	3,284.7	2,891.2	2,487.9	2,104.6
Energy Storage Systems	2,891.4	2,367.1	1,923.6	1,587.4	1,248.3
Grid Integration Services	854.6	742.8	651.9	563.2	482.7
Renewable Project Development	688.7	601.2	524.3	468.5	387.1
Total net sales	8,247.2	6,995.8	5,991.0	5,107.0	4,222.7

(b) Operating Income by Reportable Segment

Operating Income (USD millions)	2024	2023	2022	2021	2020
Photovoltaic Manufacturing	451.2	362.4	297.6	234.8	178.4
Energy Storage Systems	312.8	238.7	187.5	142.3	98.2
Grid Integration Services	138.2	114.6	94.7	78.9	61.4
Renewable Project Development	101.5	82.9	67.4	58.2	43.6
Corporate / Other	(55.4)	(47.8)	(41.5)	(38.7)	(32.1)
Total operating income	948.3	750.8	605.7	475.5	349.5

(c) Depreciation and Amortization by Reportable Segment

Depreciation & Amortization (USD millions)	2024	2023	2022	2021	2020
Photovoltaic Manufacturing	278.6	234.7	198.4	168.7	142.3
Energy Storage Systems	191.8	156.3	128.5	104.2	87.9
Grid Integration Services	51.3	44.2	38.6	32.7	28.4
Renewable Project Development	79.2	69.8	61.4	53.8	46.7
Corporate / Other	20.6	18.9	17.4	16.1	14.2
Total depreciation & amortization	621.5	523.9	444.3	375.5	319.5

(d) Capital Expenditures by Reportable Segment

Capital Expenditures (USD millions)	2024	2023	2022	2021	2020
Photovoltaic Manufacturing	562.1	487.3	423.8	385.6	298.7
Energy Storage Systems	401.7	312.5	245.1	198.4	156.2
Grid Integration Services	112.9	98.4	84.6	71.3	62.8
Renewable Project Development	168.4	147.9	134.2	121.7	103.5
Corporate / Other	27.1	24.6	22.8	21.4	18.9
Total segment capital expenditures	1,272.2	1,070.7	910.5	798.4	640.1

Segment capital expenditures presented in the table above include amounts directly attributable to each reportable segment and Corporate / Other. They differ from the consolidated 'Purchases of property, plant and equipment' line of \$1,284.9 million (2024), \$1,082.4 million (2023), \$921.4 million (2022), \$812.7 million (2021), and \$651.1 million (2020) shown in the Consolidated Statements of Cash Flows due to timing of cash payments to vendors. Segment capital expenditures also exclude capitalized interest and certain centrally managed software development costs, totaling \$30.9 million in 2024, \$24.6 million in 2023, \$23.6 million in 2022, \$16.7 million in 2021, and \$21.4 million in 2020, which are reflected in 'Additions to property, plant and equipment' in Note 7.

(e) Total Assets by Reportable Segment

Total Assets (USD millions)	Dec 31, 2024	Dec 31, 2023
Photovoltaic Manufacturing	3,412.6	2,987.4
Energy Storage Systems	2,218.7	1,843.2
Grid Integration Services	478.3	421.8
Renewable Project Development	698.1	612.9
Corporate / Other	962.3	894.7
Total assets	7,770.0	6,760.0

Note 15. Geographic Information

The following table presents net sales by country in which the customer is located. No customer accounted for more than 10% of consolidated net sales in any period presented.

Net Sales (USD millions)	2024	2023	2022	2021	2020
United States	3,198.4	2,742.6	2,387.4	2,098.7	1,768.4
Germany	1,087.2	941.7	812.5	698.2	587.3
China	1,342.6	1,187.5	1,023.8	892.4	734.9
Japan	548.9	487.2	421.7	368.4	312.8
United Kingdom	421.3	368.9	324.8	287.4	248.6
Brazil	329.8	287.4	248.7	221.6	187.4
United Arab Emirates	232.7	201.5	174.2	148.3	124.7
Other countries	1,086.3	779.0	597.9	392.0	258.6
Total net sales	8,247.2	6,995.8	5,991.0	5,107.0	4,222.7

Long-lived assets (property, plant and equipment, net) located in the United States were \$2,847.6 million at December 31, 2024 (2023: \$2,431.2 million). Long-lived assets located in Germany were \$612.4 million (2023: \$548.7 million), and the remainder were located in other foreign jurisdictions.

Note 16. Selected Quarterly Financial Data (Unaudited)

(USD millions, except EPS)	Q1 2024	Q2 2024	Q3 2024	Q4 2024
Net sales	1,872.4	2,041.6	2,103.8	2,229.4
Gross profit	498.4	548.7	574.2	626.1
Operating income	201.7	232.6	248.9	265.1
Net income attributable to common stockholders	131.4	152.6	162.4	175.0
Diluted EPS	1.45	1.68	1.79	1.92
Capital expenditures	284.7	312.4	324.6	363.2

(USD millions, except EPS)	Q1 2023	Q2 2023	Q3 2023	Q4 2023
Net sales	1,612.4	1,728.9	1,784.3	1,870.2
Gross profit	421.6	458.7	478.4	525.5
Operating income	168.3	187.4	197.6	225.7
Net income attributable to common stockholders	108.4	118.7	128.9	141.6
Diluted EPS	1.20	1.31	1.42	1.54
Capital expenditures	234.6	258.9	267.4	321.5

Quarterly capital expenditures shown above reflect cash paid for purchases of property, plant and equipment as presented in the Consolidated Statements of Cash Flows on a quarterly basis. Sum of quarterly amounts may not equal the annual amount due to rounding.

Note 17. Subsequent Events

On February 4, 2025, the Company's Board of Directors authorized an additional \$1.0 billion under the Company's share-repurchase program. On February 12, 2025, the Company executed a binding letter of intent to acquire Aurora Inverters, Inc., a U.S. string-inverter manufacturer, for approximately \$215.0 million in cash, subject to customary closing conditions. The acquisition, if completed, is expected to close in the third quarter of 2025 and will be reported within the Grid Integration Services segment.

Signatures

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this Report to be signed on its behalf by the undersigned, thereunto duly authorized.

/s/ Eleanor R. Marquez

Eleanor R. Marquez, Chief Executive Officer

Dated: February 21, 2025

/s/ David P. Schroeder

David P. Schroeder, Chief Financial Officer (Principal Financial and Accounting Officer)

Dated: February 21, 2025